Foundation

MASTER OVERVIEW — 25 AI FEATURES PRODUCTIVITY PLATFORM

READ THIS FIRST — Then Build Sprint by Sprint

HOW TO USE THESE FILES

This project is split into **sprint files** to save credits and reduce errors.

DO NOT try to build everything at once.

BUILD ORDER:

1. Read this overview file completely
2. Then paste Sprint 1 → build it → verify it works
3. Then paste Sprint 2 → build it → verify it works
4. Continue through Sprint 10
5. Finish with Sprint 11 (polish)

RULE: Never start a new sprint until the current one compiles and runs without errors.

EXISTING PLATFORM (ALREADY BUILT)

The following modules ALREADY exist. Do NOT rebuild them. The AI features will integrate with them.

```
EXISTING MODULES:
├── Authentication (NextAuth.js — Google OAuth, email/passwor
d, magic link)
```

```
├── Workspaces (multi-tenant, roles: owner/admin/editor/viewe
r/guest)
├── Task Management (databases with table/board/calendar/gall
ery/list/timeline views)
├── Documents (Tiptap block editor, nested pages, real-time c
ollab via Yjs)
├── Messaging (channels: public/private/DM, threads, reaction
s, files, Socket.io)
├── Meetings (scheduling, audio recording, Whisper transcript
ion)
├── Calendar (Google Calendar two-way sync)
└── Focus Tools (Pomodoro timer, focus sessions)
```

TECH STACK (MANDATORY — Do Not Change)

```
Frontend:         Next.js 14+ (App Router), TypeScript strict
mode, Tailwind CSS, Shadcn/UI
State:            Zustand + React Query (TanStack Query v5)
Backend:          Next.js API Routes + tRPC v11
Database:         PostgreSQL + Prisma ORM + pgvector extensio
n
Real-Time:        Socket.io with Redis adapter
AI:               OpenAI API (GPT-4o / GPT-4o-mini), provider
-agnostic service layer
Vector Search:    pgvector (1536-dimension embeddings)
File Storage:     Cloudflare R2 (S3-compatible)
Cache:            Redis (Upstash)
Background Jobs:  BullMQ with Redis
Deployment:       Replit Deployments
```

CODING STANDARDS (APPLY TO EVERY FILE)

```
1. TypeScript strict mode. Zero "any" types. Use proper gener
ics.
```

```
2. Every tRPC procedure: Zod input validation + error handlin
g + auth check
3. Every Prisma model: id (UUID), createdAt, updatedAt, delet
edAt (soft delete)
4. Every UI component: loading state, error state, empty stat
e, responsive
5. All AI calls go through src/lib/ai/ai-service.ts (NEVER ca
ll OpenAI directly)
6. Optimistic UI updates with React Query mutation + rollback
on error
7. Background jobs (BullMQ) for anything > 2 seconds
8. No pricing logic. No plan checks. No credit limits. All fe
atures available.
9. Use barrel exports (index.ts) for clean imports
10. Comments on every function explaining business logic
```

FILE STRUCTURE

```
src/
├── app/                          # Next.js App Router pages
│   ├── (auth)/                   # Auth pages (login, regist
er)
│   ├── (dashboard)/              # Main app layout
│   │   ├── [workspaceId]/
│   │   │   ├── ai-agents/        # Feature 4: Agent list, bu
ilder, chat
│   │   │   ├── calendar/         # Feature 16: Calendar opti
mizer
│   │   │   ├── command-center/   # Feature 23: Personal dash
board
│   │   │   ├── daily-plan/       # Feature 3: Daily planner
│   │   │   ├── documents/        # Existing + Feature 1 AI w
riting
│   │   │   ├── email/            # Feature 13: Email assista
nt
```

```
|   |   |   ├── goals/            # Feature 15: Goals & OKRs
|   |   |   ├── habits/           # Feature 14: Habit tracker
|   |   |   ├── integrations/     # Feature 24: Integration h
ub
|   |   |   ├── knowledge/        # Feature 2: Knowledge base
browser
|   |   |   ├── meetings/         # Existing + Feature 7 inte
lligence
|   |   |   ├── messaging/        # Existing + Feature 11 int
elligence
|   |   |   ├── my-day/           # Feature 3: My Day view
|   |   |   ├── projects/         # Existing + Feature 5 AI P
M
|   |   |   ├── settings/
|   |   |   |   ├── ai/           # AI settings (models, bran
d voice)
|   |   |   |   ├── integrations/ # Connected apps settings
|   |   |   |   └── workspace/    # Workspace settings
|   |   |   ├── tasks/            # Existing + Feature 6 inte
lligence
|   |   |   ├── team-insights/    # Feature 22: Team analytic
s
|   |   |   ├── templates/        # Feature 17: Template gall
ery
|   |   |   ├── time-tracking/    # Feature 21: Time analytic
s
|   |   |   └── workflows/        # Feature 19: Workflow reco
rder
|   |   └── layout.tsx            # Dashboard layout with sid
ebar
|   ├── api/
|   |   └── trpc/[trpc]/route.ts  # tRPC API handler
|   └── layout.tsx                # Root layout
|
├── server/                       # Backend logic
|   ├── routers/                  # tRPC routers (one per fea
```

```
ture)
│   │   ├── ai-writing.ts        # Feature 1
│   │   ├── knowledge.ts         # Feature 2
│   │   ├── daily-plan.ts        # Feature 3
│   │   ├── agents.ts            # Feature 4
│   │   ├── project-manager.ts   # Feature 5
│   │   ├── task-intelligence.ts # Feature 6
│   │   ├── meetings.ts          # Feature 7
│   │   ├── database-ai.ts       # Feature 8
│   │   ├── focus.ts             # Feature 9
│   │   ├── doc-generator.ts     # Feature 10
│   │   ├── messaging-ai.ts      # Feature 11
│   │   ├── ai-engine.ts         # Feature 12
│   │   ├── email.ts             # Feature 13
│   │   ├── habits.ts            # Feature 14
│   │   ├── goals.ts             # Feature 15
│   │   ├── calendar-optimizer.ts# Feature 16
│   │   ├── templates.ts         # Feature 17
│   │   ├── voice.ts             # Feature 18
│   │   ├── workflows.ts         # Feature 19
│   │   ├── notifications.ts     # Feature 20
│   │   ├── time-tracking.ts     # Feature 21
│   │   ├── team-insights.ts     # Feature 22
│   │   ├── command-center.ts    # Feature 23
│   │   ├── integrations.ts      # Feature 24
│   │   ├── onboarding.ts        # Feature 25
│   │   └── _app.ts              # Root router merging all ro
uters
│   └── services/                 # Business logic services
│       ├── ai-service.ts        # Feature 12: Multi-model AI
engine
│       ├── knowledge-indexer.ts  # Feature 2: Indexing pipel
ine
│       ├── plan-generator.ts    # Feature 3: Daily plan algo
rithm
│       ├── agent-executor.ts    # Feature 4: Agent execution
```

```
engine
│       ├── risk-detector.ts     # Feature 5: Risk detection
│       └── calendar-engine.ts   # Feature 16: Calendar optim
ization
│
├── lib/                          # Shared utilities
│   ├── ai/
│   │   ├── ai-service.ts        # Central AI service (provid
er-agnostic)
│   │   ├── providers/
│   │   │   ├── openai.ts        # OpenAI provider
│   │   │   ├── anthropic.ts     # Anthropic provider
│   │   │   └── gemini.ts        # Google Gemini provider
│   │   ├── prompts/             # System prompts per feature
│   │   │   ├── writing.ts
│   │   │   ├── task-breakdown.ts
│   │   │   ├── meeting-summary.ts
│   │   │   └── ...
│   │   └── embeddings.ts        # Embedding generation + cac
hing
│   ├── prisma.ts                # Prisma client singleton
│   ├── redis.ts                 # Redis client singleton
│   ├── trpc.ts                  # tRPC config
│   └── utils.ts                 # Shared helpers
│
├── components/                   # Reusable UI components
│   ├── ai/                      # AI-specific components
│   │   ├── ai-prompt-bar.tsx    # Cmd+J prompt bar (used eve
rywhere)
│   │   ├── ai-streaming-display.tsx  # Streaming text displa
y
│   │   ├── ai-diff-view.tsx     # Before/after text comparis
on
│   │   ├── ai-suggestion-card.tsx    # Suggestion popover
│   │   ├── ai-model-selector.tsx     # Model dropdown
│   │   └── ai-loading.tsx       # AI processing spinner
```

```
│   ├── daily-plan/
│   │   ├── plan-timeline.tsx    # Feature 3: Timeline UI
│   │   ├── plan-block.tsx       # Feature 3: Single time blo
ck
│   │   └── morning-ritual.tsx   # Feature 3: Planning flow
│   ├── agents/
│   │   ├── agent-chat.tsx       # Feature 4: Chat interface
│   │   ├── agent-builder.tsx    # Feature 4: Create/edit age
nt
│   │   └── agent-card.tsx       # Feature 4: Agent list card
│   └── ...                      # More per feature
│
├── hooks/                        # Custom React hooks
│   ├── use-ai-stream.ts         # Hook for streaming AI resp
onses
│   ├── use-keyboard-shortcut.ts # Keyboard shortcut handler
│   └── use-focus-session.ts     # Focus session state
│
├── stores/                       # Zustand stores
│   ├── ai-store.ts              # AI state (current model, l
oading)
│   ├── daily-plan-store.ts      # Daily plan state
│   ├── focus-store.ts           # Focus session state
│   └── notification-store.ts    # Notification state
│
├── jobs/                         # BullMQ background jobs
│   ├── index-knowledge.ts       # Feature 2: Index pages/tas
ks/messages
│   ├── generate-daily-plan.ts   # Feature 3: Generate daily
plan
│   ├── execute-agent.ts         # Feature 4: Run agent
│   ├── detect-risks.ts          # Feature 5: Risk detection
scan
│   ├── process-meeting.ts       # Feature 7: Transcribe + su
mmarize
│   ├── sync-calendar.ts         # Feature 16: Calendar sync
```

```
│   ├── send-notifications.ts    # Feature 20: Batch notifica
tions
│   └── queue-config.ts          # Queue definitions + retry
config
│
└── prisma/
    └── schema.prisma            # ALL database models (built
in Sprint 1)
```

ALL 25 FEATURES — SUMMARY TABLE

#	Feature	Sprint	Description
1	AI Writing Assistant	2	Context-aware writing in editor + messaging
2	AI Knowledge Manager & Search	2	Universal search + natural language Q&A with citations
3	AI Auto-Scheduler & Daily Planner	3	Auto-schedule tasks into optimized daily plan
4	AI Autonomous Agents	4	24/7 AI assistants with triggers and tool-calling
5	AI Project Manager	5	Standups, risk detection, status reports, automations
6	AI Task Intelligence	5	Task breakdown, prioritization, NLP input, duration estimation
7	AI Meeting Intelligence	6	Recording, transcription, summaries, action items, prep
8	AI Database Intelligence	5	Auto-fill columns, NL queries, AI database creation
9	AI Focus & Productivity Coach	3	Focus protection, sessions, analytics, coaching
10	AI Document Generator	6	Generate reports from workspace data
11	AI Messaging Intelligence	7	Summarize, catch-up, smart replies, translate, decisions

#	Feature	Sprint	Description
12	Multi-Model AI Engine	1	GPT-4o + Claude + Gemini, auto-routing, centralized service
13	AI Email Assistant	7	Compose, triage, follow-ups, smart labels
14	AI Habit Tracker & Streak Engine	3	Recurring habits, AI scheduling, streaks
15	AI Goal & OKR System	8	Hierarchical goals, auto-progress tracking, risk alerts
16	AI Calendar Optimizer	8	Meeting defrag, focus protection, team scheduling
17	AI Template Engine	9	AI template generation, gallery, recommendations
18	AI Voice Commands & Notes	9	Voice input, transcription, commands
19	AI Workflow Recorder	4	Record workflows, AI cleans up, replay forever
20	AI Smart Notifications	9	AI triage, priority inbox, batching
21	AI Time Tracking	8	Auto-tracking, estimation, analytics
22	AI Team Insights	10	Workload, focus, burnout detection
23	AI Personal Command Center	10	Unified dashboard aggregating all features
24	AI Integration Hub	10	Connect Google Drive, GitHub, Jira, Slack
25	AI Onboarding Assistant	10	Interactive walkthroughs, training chatbot

SPRINT DEPENDENCIES (NEVER SKIP)

```
Sprint 1 (Foundation) → Required by ALL other sprints
Sprint 2 (Writing + Knowledge) → Required by Sprint 4 (Agent
s), Sprint 5 (PM), Sprint 6 (Meetings)
Sprint 3 (Planner + Habits + Focus) → Required by Sprint 8 (C
alendar), Sprint 10 (Dashboard)
Sprint 4 (Agents + Workflows) → Required by Sprint 5 (PM uses
agents)
Sprint 5 (PM + Tasks + Database) → Required by Sprint 6 (Meet
ings create tasks)
Sprint 6 (Meetings + Documents) → Can proceed independently a
fter Sprint 5
Sprint 7 (Messaging + Email) → Can proceed independently afte
r Sprint 2
Sprint 8 (Goals + Calendar + Time) → Requires Sprint 3 + Spri
nt 5
Sprint 9 (Notifications + Templates + Voice) → Requires Sprin
t 2 + Sprint 3
Sprint 10 (Dashboard + Team + Integrations + Onboarding) → Re
quires most prior sprints
Sprint 11 (Polish) → Final pass after all features built
```

CROSS-FEATURE DATA FLOW

```
Feature 12 (AI Engine) ──powers──→ ALL other AI features
Feature 2 (Knowledge) ──indexes──→ Pages, Tasks, Messages, Me
etings, Files
Feature 3 (Daily Plan) ──uses──→ Tasks (Feature 6), Habits (F
eature 14), Calendar (Feature 16)
Feature 4 (Agents) ──uses──→ Knowledge (Feature 2), Writing
(Feature 1), Tasks (Feature 6)
Feature 5 (Project Manager) ──uses──→ Tasks (Feature 6), Know
ledge (Feature 2), Team Insights (Feature 22)
Feature 7 (Meetings) ──feeds──→ Knowledge (Feature 2), Tasks
(Feature 6), Messaging (Feature 11)
```

```
Feature 9 (Focus) ──integrates──→ Daily Plan (Feature 3), Cal
endar (Feature 16), Notifications (Feature 20)
Feature 15 (Goals) ──tracks──→ Tasks (Feature 6), Habits (Fea
ture 14)
Feature 23 (Command Center) ──aggregates──→ Daily Plan, Habit
s, Goals, Notifications, Meetings, Tasks
```

READY TO START

Now paste the Sprint 1 file into Replit AI Agent.
Sprint 1 builds: ALL database schemas + AI service layer + shared UI components.
Without Sprint 1, nothing else works.